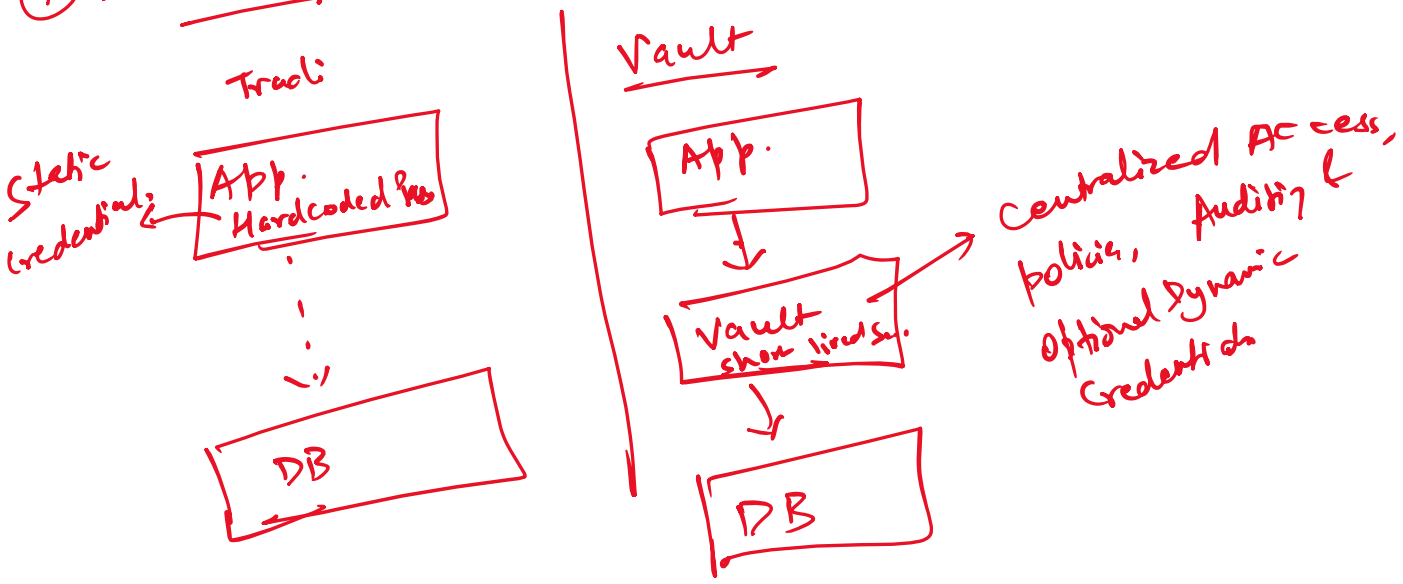
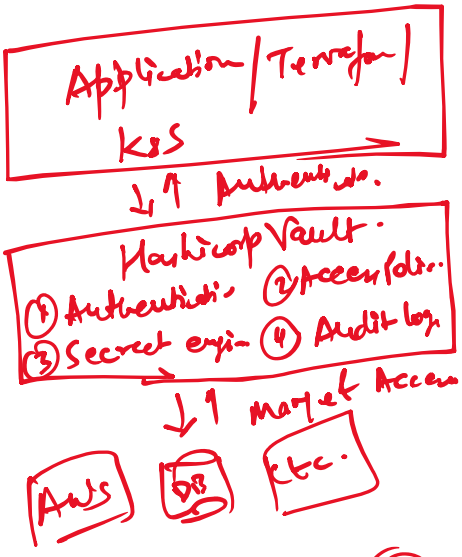
① Secret Mgt :- Secure storage, accessing.



② Terraform Vault Provider :- Why?

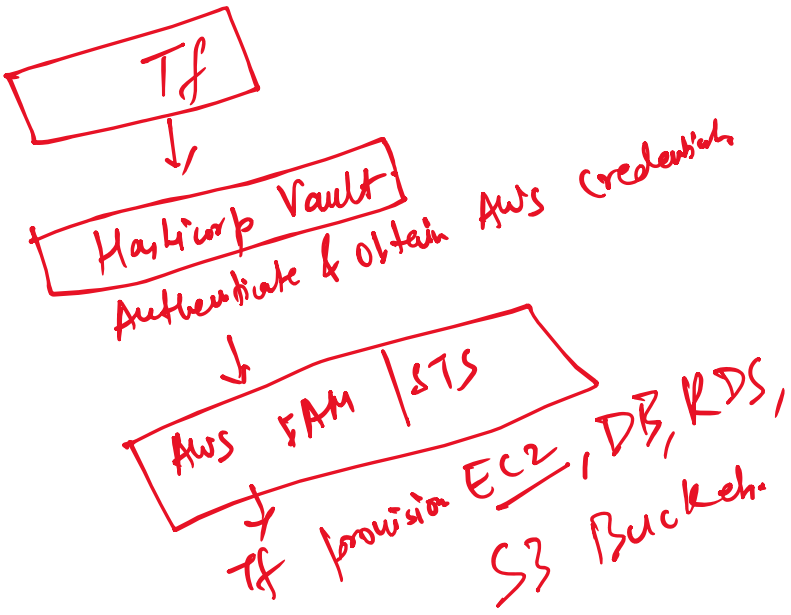


① Secret Engine.
② Authentication Method.
③ Policies.
④ Token

⑤ Lease / TTL :-
⑥ Audit Devices :-
⑦ Transit Engine :-

Static & Dynamic secrets :-

Static & Dynamic secrets:-

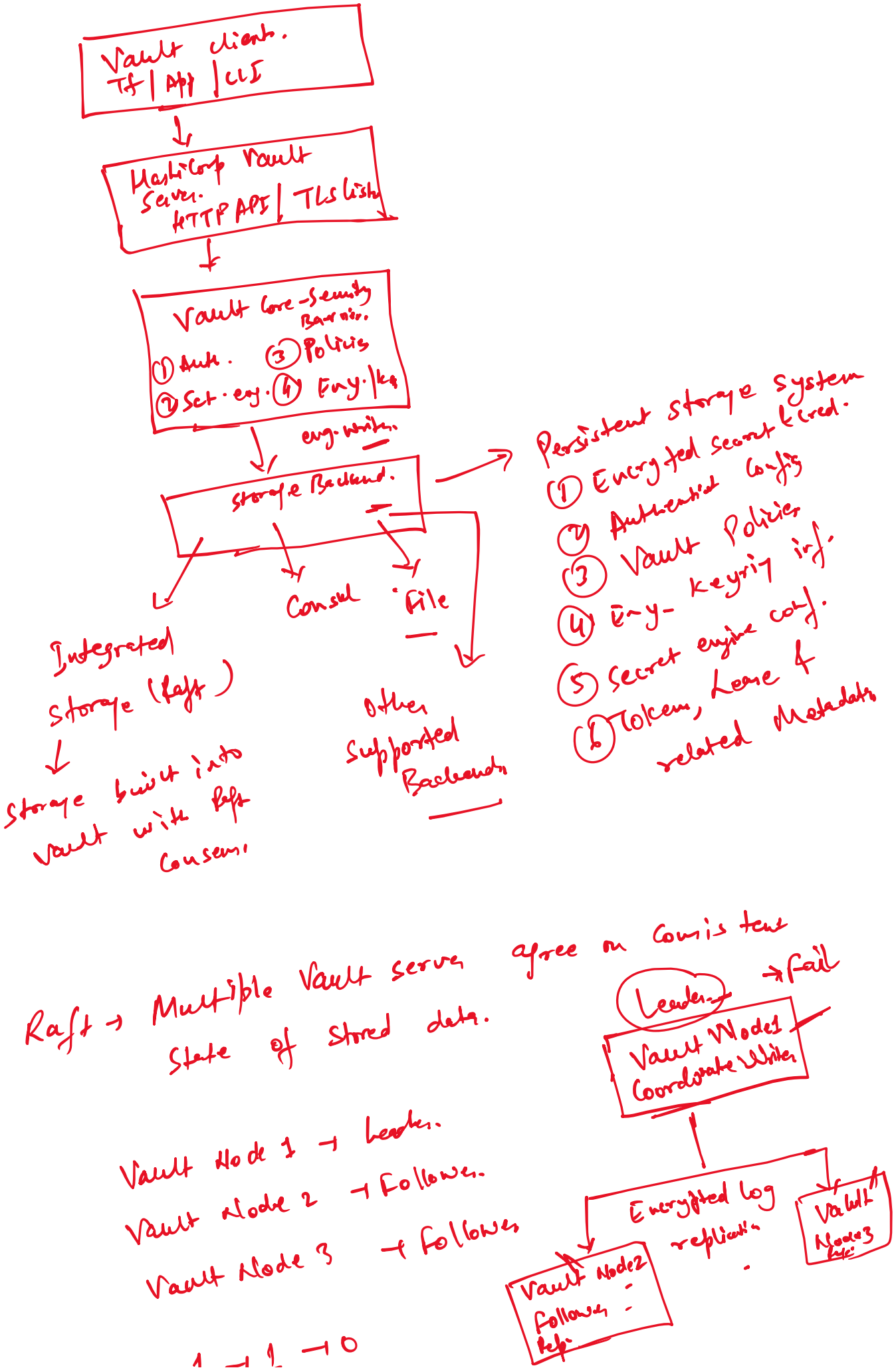


1. Vault status.
2. Vault secret.
3. Vault kv put.
4. Vault kv get.
5. Vault kv delete

6. Vault Secret enable.

2. Vault Architecture

1. Storage Backend. → where?
2. Secret Barrier → How?
3. Seal | unseal. → How Vault lock & Unlock Access for encrypted data.
4. Shamir's Secret sharing → Multiple Administrator jointly control Unseal process.



Raft → Multiple Vault servers agree on consistent state of stored data.

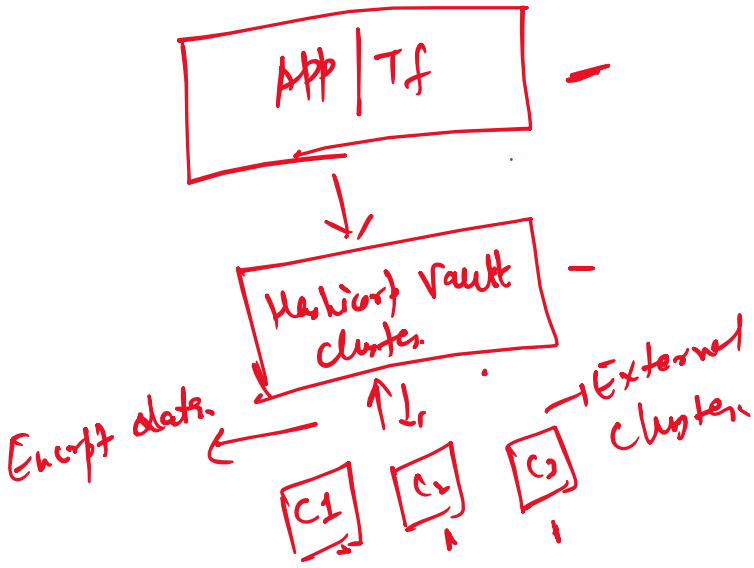
Vault Node 1 → Leader.
Vault Node 2 → Follower.
Vault Node 3 → follower

1 → 1 → 0

Follower - Rep.

1 → 1 → 0
3 → 2 → 1
5 → 3 → 2
7 → 4 → 3

Hashicorp Consul ⇒



Consul

1. External Cluster
2. Raft
3. Add. Infra.
4. Higher.
5. Existing.

Vault Integrated Storage.

1. Own storage.
2. Raft
3. No separate Infra
4. Generally lower.
5. New vault.

(1) Simple Arch:

With Consul.

App.
↓

With Integrated storage

App.
↓
Vault cluster

App.
↓
Vault Cluster.
↓
Consull Cluster.
↓
Persistent storage

↓
Vault cluster
↓
Integrated Raft storage.

(2.) Fewer Infra. Dependencies

(3.) Easier Maintenance

(4) Cost efficient.

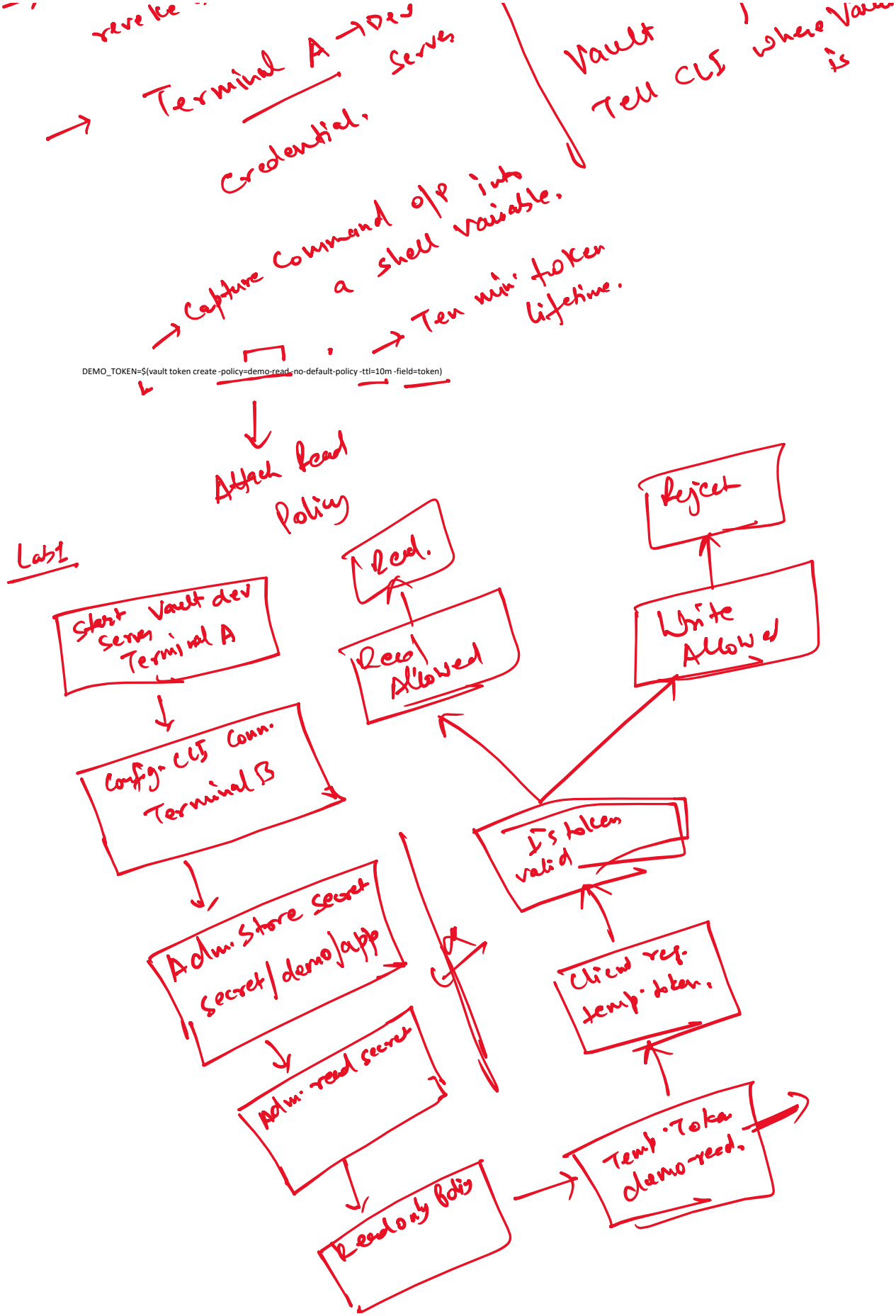
Install Vault
↓
Lab1:- Temp. Dev Server.
↳ stop Dev server ; its data disappear.
↓
Lab2: persistent Vault server
↓
Lab3: Restart & Verify persistence.
↓
Lab4: CLI, API, UI
↓
Lab5: Token, Userpass, LDAP
↓
Lab6: KV Versioning & Recovery.

(1.) Intro. Secret Mng. & Vault.

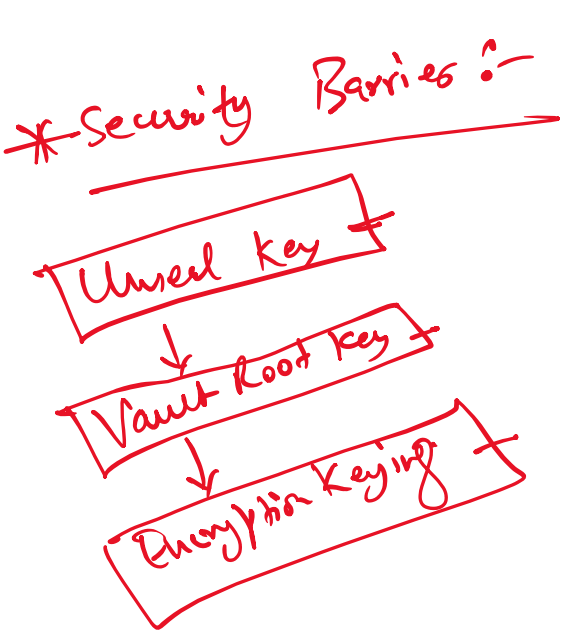
→ Store an app. secret centrally, allow read only access! revoke that access.

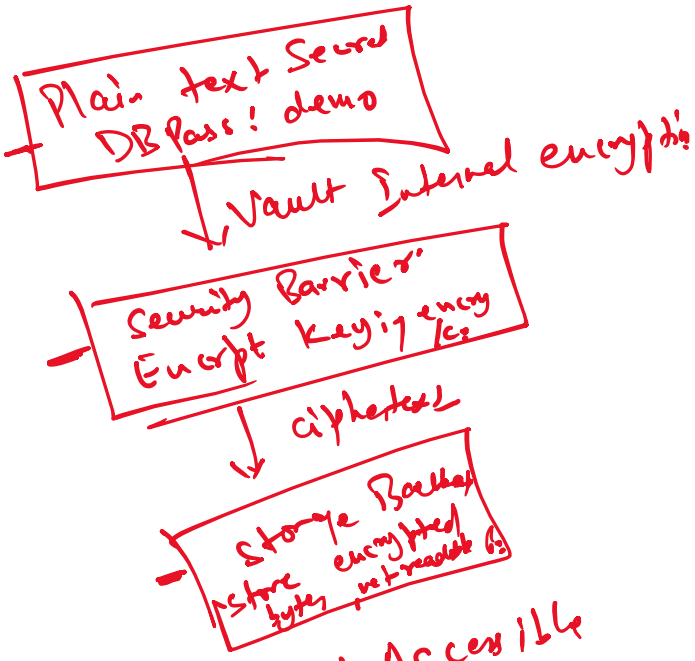
A → Dev Server

Terminal B → Vault ↑ where Vault is


Terminal A → Dev Server
Credential.
Vault
Tell CLI where Vault is
→ Capture Command o/p into a shell variable.
→ Ten min. token lifetime.
DEMO_TOKEN=$(vault token create -policy=demo-read-no-default-policy -ttl=10m -field=token)
Attach read Policy
Lab1
Start Vault dev Server Terminal A
Config. CLI Conn. Terminal B
Adm. store secret secret/demo/app
Adm. read secret
Read only Policy
Is token valid
Client req. temp. token.
Temp. Token demo-read.
Read
Read Allowed
Reject
Write Allowed

* Security Barrier :-





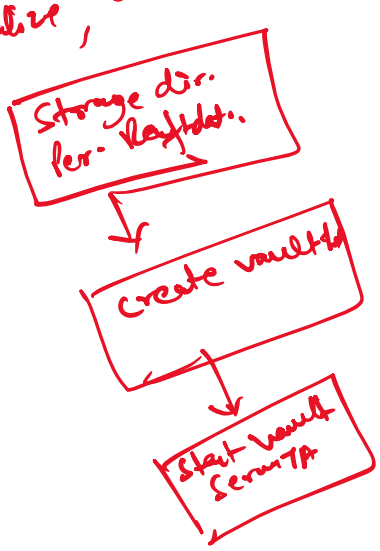
(1) Vault Seal & Unseal → Keying Accessible

Process locking Vault

ability to decrypt

Persisted data.

* Initialize, unseal, , store & seal vaults



1 share - 1 threshold → lower.

3 share - 2 thr → med.

5 share - 3 → strong.

7 share - 5 thr → high.

10 share - 7 thr - high.

Authorized write

↓

KV secret engine

↓ data.

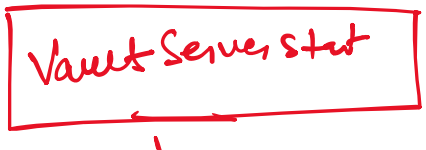
KV secret engine

↓

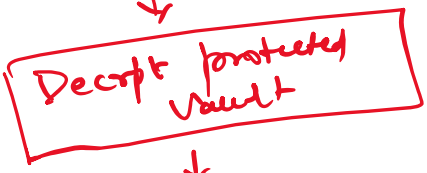
Security barrier encrypts data.

↓

Raft persists encrypted data on disk



↓



↓



* Dev Mode vs Persistent Server Mode :-

HCL Vault

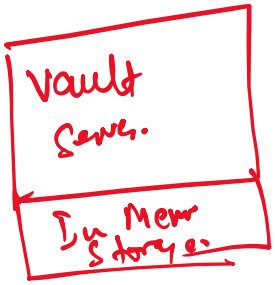
- Dev. → Vault server -dev → In Mem
- Persistent

Vault server -config = vault.hcl.

↓

Raft | consul | other configs Backend

Dev Mode



↓

Restart → Data lost

Per.



↓

Restart → Data Retained.

Authentication Method → Token, Userpass, LDAP

Authentication → Who are you?

allowed to do?

Authentication → Who are you?

Authorization → What are you allowed to do?

1. Token Authentication. → Vault token

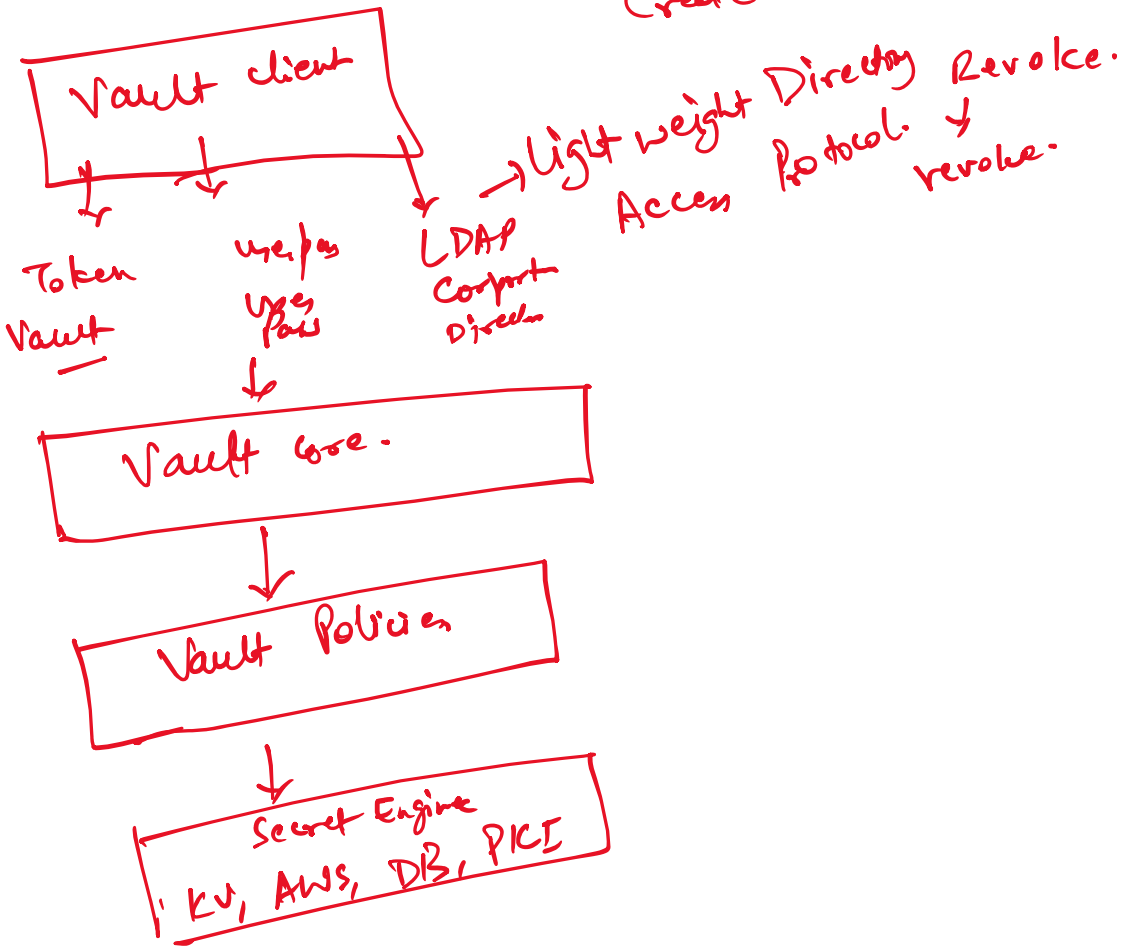
2. Userpass Authentication. → username & password.

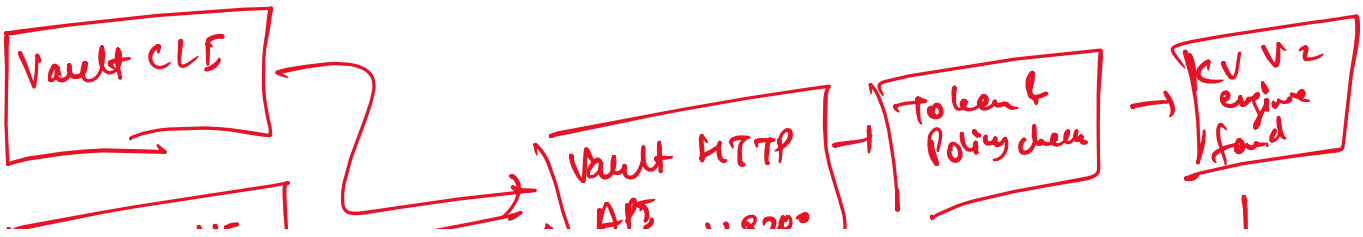
3. LDAP Authentication → Directory → Microsoft AD

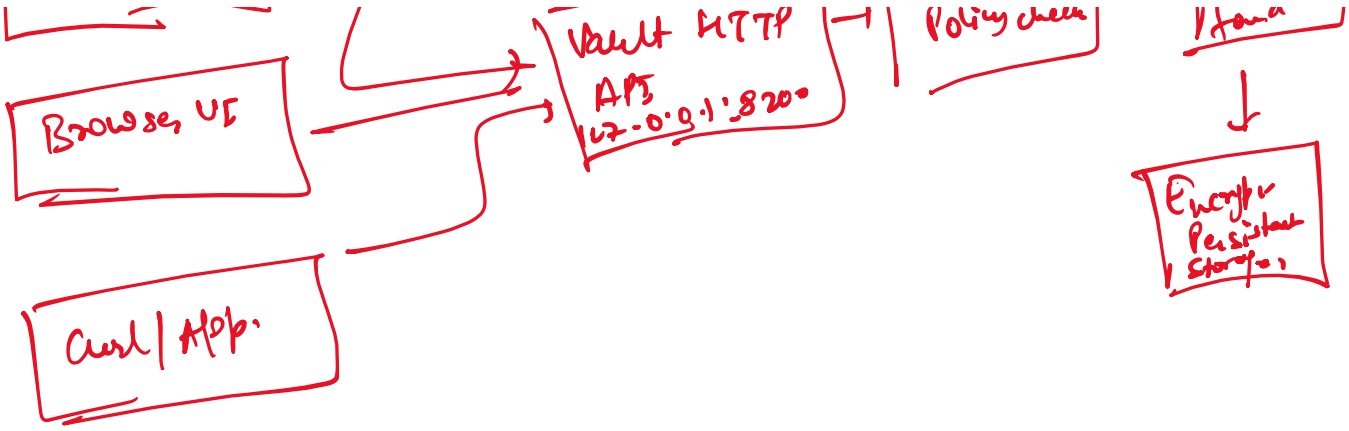
Vault token create → Create → Inspect → Renew

lookup

renew






Browser UI
Vault HTTP API 127.0.0.1:8200
Policy check
Home
Encrypted Persistent storage
Curl / App.